Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, R, SQL, Java, C/C++
Data Science & Machine Learning: Predictive Modelling, Classification, Ensemble Methods, Probabilistic Modelling, Time-Series Analysis, Anomaly Detection, Feature Engineering, Model Evaluation, Sentiment Analysis
Financial Analytics: Financial Data Analysis, Risk Scoring, Transaction Analytics, Market Data Analysis, EDA
Generative AI & NLP: LLM Applications, RAG, Prompt Engineering, NLP, TF-IDF Retrieval
Data Science Tools: Pandas, NumPy, Scikit-learn, PyTorch, Power BI, DAX, Streamlit, Tableau, Matplotlib
Software Engineering: DSA, OOPS, Operating Systems, DBMS, Multithreading & Concurrency, REST APIs, Unit Testing, TDD
Tools & Platforms: Git, GitHub, Jupyter Notebook, FastAPI, Docker, MySQL, SQLite, Redis

EXPERIENCE

VOIS – Vodafone Idea Foundation* | LINK** — ***Aug – Sep, 2025

- Analysed 2 real-world business datasets containing 8,500+ records, applying data preprocessing and AI-assisted analytics to identify trends and generate 15 actionable business insights for data-driven decision-making.
- Validated analytical findings for accuracy and consistency, reviewing outputs to ensure reliable and meaningful insights for business recommendations.
- Established a standardised analytics workflow covering data preprocessing, insight generation, and validation, enabling consistent conversion of raw business data into actionable recommendations.

PROJECTS

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered an end-to-end financial analytics pipeline processing 50K+ PaySim transactions, combining statistical outlier analysis, duplicate detection, vendor-velocity checks, HHI concentration analysis, and Isolation Forest for transaction-risk detection.
- Designed a star-schema SQLite data warehouse and composite 0–100 risk scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Built a 5-page Power BI audit dashboard with 20+ DAX measures to surface vendor risk, anomaly rates, transaction trends, and risk insights, supported by automated pipeline tests.

***RedisLite* | LINK**

- Engineered a Redis-inspired in-memory key-value store in Python, implementing a custom RESP protocol parser and multithreaded TCP server for concurrent client requests.
- Implemented Write-Ahead Logging, crash recovery, thread-safe storage, and LRU caching, combining systems programming with performance-oriented data structures.
- Validated reliability through 46 automated tests and benchmarked 619K SET operations/sec, demonstrating correctness and high-throughput request handling.

***Docwise* | LINK**

- Engineered an end-to-end RAG-based document analytics pipeline using Google Gemini, NLP, TF-IDF retrieval, and intelligent chunking, achieving 95%+ query accuracy on PDFs exceeding 200 pages.
- Developed the complete document intelligence workflow from document extraction and chunking to retrieval, contextual ranking, and LLM-based response generation.
- Improved response reliability through retrieval tuning, prompt engineering, and hallucination-aware validation, increasing contextual relevance and answer consistency.

***Real-Time Stock Market Anomaly Detector* | LINK**

- Engineered a real-time market analytics pipeline processing 2M+ NSE records, combining price signals and financial-news sentiment to detect unusual market behaviour.
- Developed and evaluated Isolation Forest and LSTM Autoencoder models, achieving 88% precision and 85% recall on validation data for anomaly detection.
- Optimised the analytics pipeline to reduce detection latency by 35%, enabling near real-time monitoring and faster identification of unusual market events.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | LINK — 2026
- Deloitte Australia Data Analytics Job Simulation | LINK — 2025